

POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

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Jaypee Institute of Information Technology, Noida

Test 1 Examination, ODD 2023

B.Tech(CSE/IT/ECE)/Dual, I Semester

Course Title: Software Development Fundamentals-1 Time: 1 hour

Course Code: 15B11CI111

Maximum Marks: 20

COs	Course Outcomes
C109.1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithm and flowcharts.
C109.2	Explain basics of C programming concepts to make decision for solving problems.
C109.3	Demonstrate and contrast different methods for writing modular programs in C
C109.4	Use various C programming constructs to implement iteration, and recursion
C109.5	Apply and implement arrays, pointers, structures and file handling for solving real world problems

Attempt all questions

Q1. Four people (Alex, Bob, Carol, and Dave) come across a bridge at night. The bridge can only hold two people at a time, and they only have one flashlight (crossing the bridge without flashlight is impossible). The four people walk at different speeds: Alex takes 1 minute to cross the bridge, Bob takes 2 minutes, Carol takes 5 minutes, and Dave takes 10 minutes. When two people cross the bridge together, they must go at the slower person's pace. They have only 17 minutes before a train passes under the bridge, and they need to get to the other side before that happens. Your task is to find the shortest time in which all four people can safely cross the bridge. Explain your steps. [C109.1 (Understanding), 2 Marks]

Q2. You have been asked to design an automated face recognition system. However, the exact requirements are not yet clear. Can we use waterfall SDLC model for this particular system? Justify your answer. [C109.1 (Understanding), 1 Marks]

Q3. Write a program to input a positive integer from user. Then exchange its i^{th} and j^{th} digits. The value of i and j should be given as input by the user. Please note all the digits in the number are supposed to be non-zero.

Example: $N = 3246159$

$i = 2, j = 5$

i^{th} digit is 5 and j^{th} digit is 4

Exchanged number is: 3256149

[C109.4 (Applying), 5 Marks]

Q4. Write a C program to automate the process of apparel label generation for Men's trousers. The program should take inputs of standard Hip and Length in inches. The program should identify the standard apparel Label Size based on the table using Switch-case control as given below.

Hip (in inches)	Length (in inches)	Label Size
30	38	LARGE
32	39	XL
34	40	XXL

Finally, the output should display the appropriate label of Hip, Length, and Label Size. Sample label format generated is given as below.

Hip : 32
Length : 39
Label Size : XL

For any other input display the message "Wrong Input".

[C109.2 (Understanding), 3 Marks]

Q5. You are asked to draw a flowchart that determines a person's income tax based on their income and other criteria. The tax calculation is based on the following rules:

Income	Tax
Less than 700000	0
Greater than 700000 and less than equal to 1000000	10%
Greater than 1000000 and less than equal to 1500000	30000 + 20% of amount exceeding 1500000
Greater than 1500000	130000 + 30% of amount exceeding 1500000

Additionally, if the total tax of an individual exceeds 30,000 then an additional 5% education cess tax is applied on their tax.

[C109.1 (Understanding), 3 Marks]

Q6. Write a C program to print a butterfly pattern based on the height of the butterfly provided by the user. Your program must ensure that the height entered by the user must be an even positive integer.

Enter height: 10

[C109.4 (Applying), 6 Marks]

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